

Random and Systematic Error

Making sense of scatter in repeated measurements

Physics · 1A · 23 September 2026

The stopwatch has a resolution too

(slide 2)

A digital stopwatch usually reads to 0.01 s. That is its **sensitivity** (resolution): the smallest change it can show. For an analogue scale the rule is different: you read to half of the smallest division. For a digital display the instrument gives you 0.01 s, but your own reaction time is usually the real limit, not the display.

Before moving on. Is 0.01 s really how precise your reading is?

We do not answer yet. The answer will arrive on its own when we see many different readings of exactly the same event.

Two different kinds of error

(slide 3)

- **Random error.** Unpredictable: it scatters the values above and below the true value. It is reduced by repeating the measurement and averaging.
- **Systematic error.** Consistent: it shifts every reading the same way, for example a zero error. Repeating the measurement does not remove it.

A classic trap: thinking that more repeats fix everything. Repeating helps only with random error. The real work of telling the two apart happens later, with real data.

Four dart boards

(slide 4)

Picture four dart boards (figure 1). The bullseye is the true value.

1. **Accurate and precise:** a tight cluster on the bullseye. This is the goal.
2. **Precise, not accurate:** a tight cluster, but off-centre. The offset is a *systematic* error; note that a tight, confident-looking cluster can still be completely wrong.
3. **Accurate, not precise:** widely scattered, but centred on the bullseye. This is *random* error. The average of a scattered-but-centred set is still correct, which is exactly why averaging helps with random error and not with systematic error: the average of board 2 is still off-centre.
4. **Neither:** scattered and off-centre, both errors together.

We will come back to this picture later in the lesson.

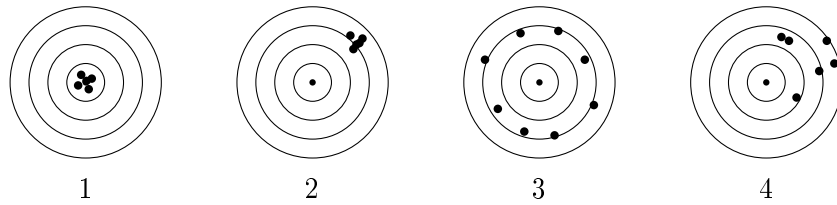


Figure 1: Four dart boards; the bullseye is the true value. Board 1: accurate and precise. Board 2: precise, not accurate. Board 3: accurate, not precise. Board 4: neither.

Describing a set of repeated readings

(slide 5–6)

For a set of repeated measurements:

$$\text{mean} = \frac{\text{sum of readings}}{\text{number of readings}} \quad \text{range} = \text{highest} - \text{lowest}$$

Today we use only mean and range: no median, no mode. They are not part of what Paper 6 asks.

Figure 2 makes the two formulas visual with a toy example, not today's real data: six readings sit on the number line, the mean is the dashed line, and the range is the bracket between the lowest and the highest value.

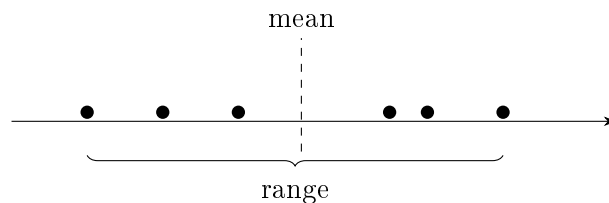


Figure 2: Six readings on a number line, with their mean (dashed line) and their range (bracket). Illustrative positions, not real data.

The swinging apple

(slide 7–9)

Now the live demonstration: one swing, one event, 28 stopwatches. A model apple on a string makes one complete swing, out and back (figure 3). Before it swings, each of you writes down your own prediction of the time, on paper. Then everyone times the same single swing, independently, on their laptop. We use laptops, not physical stopwatches, so that no two students read the same display: the readings must be independent.

First we agree on the event. In the animation one swing lasts 2.5 s.

Before moving on. When do you press stop?

Start is the instant the apple is released at one side; stop is the instant it comes back to that same side. Out and back, not just across: if you stop when it reaches the other side, you have timed half a swing. If we did not agree on the event, the 28 readings would measure different things and the spread would mean nothing.

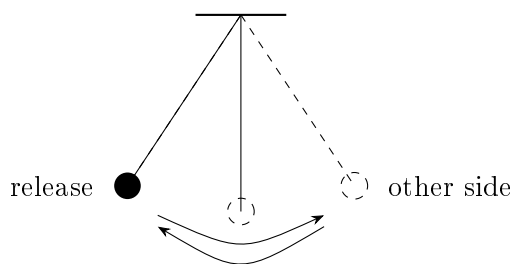


Figure 3: One complete swing: out and back to the release point (angle exaggerated).

Predict the spread

(slide 10)

Before moving on. How much will the 28 readings differ? Less than 0.2 s apart, between 0.2 s and 0.5 s apart, or more than 0.5 s apart? Write down your choice: we revisit it once the real data are in.

Data collection

(slide 11–13)

Time it now, independently, with no comparing. Start your stopwatch app at the release and stop when the apple is back at the release point. Write your value on the board when called, and do not change it after seeing other people's values: this protects the independence of the 28 readings, which is the point of the exercise. The apple is released only once: a second attempt would time a different event, not a repeat of the same one.

We have 28 readings of the same swing, not 28 separate swings. Compute their mean and range; the values are in your notebook.

Now compare your predicted spread with the real range. For a single event timed by hand the spread is very often well over 0.5 s, which usually surprises people. Even a simple stopwatch reading is limited far more by human reaction time than by the 0.01 s of its display. So 0.01 s is the sensitivity of the instrument, not the precision of your reading.

Does any value look strange?

(slide 14–15)

Plot the readings on a graph with the student number on the horizontal axis: one point per student. There is no line of best fit here, because there is no relationship to fit; best fit comes on 7 October.

Before moving on. Does any value look strange? Which one?

The odd one out has a name: an **anomalous result**.

Discard it, or keep it?

(slide 16)

We cannot re-measure: the swing already happened, once. So should we discard the anomalous result? The example below uses an example dataset, not the real class data: 27 readings between 2.36 s and 2.59 s, plus one reading at 3.40 s.

	Mean / s	Range / s
including the anomaly	2.52	1.04
excluding the anomaly	2.49	0.23

The anomaly barely moves the mean but wrecks the range. The conclusion is to exclude it, but only with a stated reason (for example, “that student started late”), and saying that you did so. Never delete it silently: it stays in the table, crossed out. With a repeatable experiment, such as the pendulum on 28 October, you would simply time it again.

Where does the spread come from?

(slide 17)

Suppose the true swing time is 2.50 s. We run four cases, in order.

Before moving on. For each case, what does the stopwatch read? (1) Perfect reactions. (2) You stop 0.30 s too late. (3) You stop 0.20 s too early, anticipating the return. (4) You are late by 0.25 s at *both* start and stop.

Case	What happens	Reading / s
1	perfect reactions	2.50
2	stop 0.30 s too late	2.80
3	stop 0.20 s too early	2.30
4	late by 0.25 s at start and stop	2.50

In case 4 the two delays cancel. The key idea: what spoils a reading is not being slow, but being slow by a *different* amount at start and stop, and that difference changes unpredictably from person to person and from try to try. That unpredictability is what makes the 28 readings scatter.

Is that scatter random or systematic?

(slide 18)

Another example dataset, in which the true value is known (2.50 s); in the real experiment we do not know it (figure 4). In the first case there is random error only: the readings pile up on both sides of the true value and the mean lands close to it. In the second case the readings are the same, but every one is 0.30 s too long, for example because of a stopwatch app that lags: the pile has exactly the same shape and width, it has just moved. The mean moves with it, so averaging does not help.

Before moving on. Looking only at our 28 numbers, can we tell whether there is a systematic error?

No. A systematic error does not change the spread, so it can only be caught by comparing with something independent: another instrument, or a known value. What the spread does tell us is that our readings contain random error. We name it now: *this scatter is random error*.

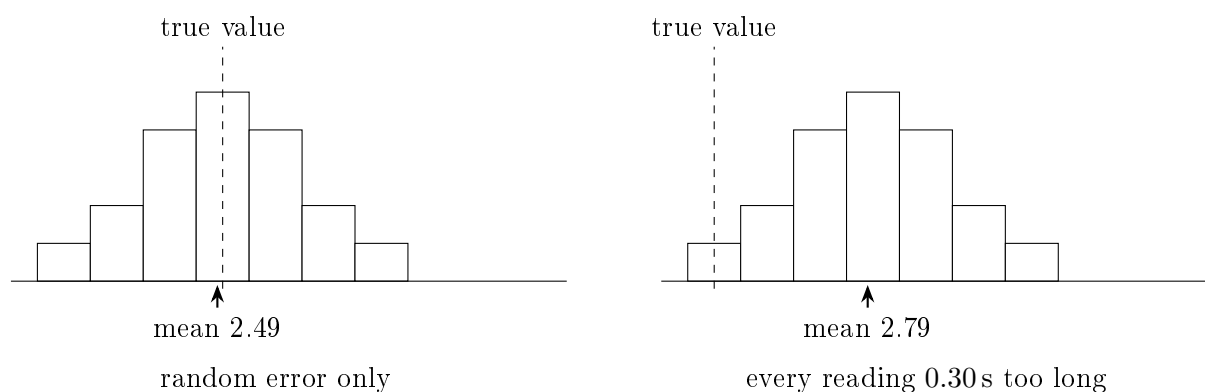


Figure 4: Schematic distribution of repeated readings. Left: random error only, the readings pile up on both sides of the true value and the mean (2.49) lies close to it. Right: the same pile shifted by 0.30 s, the shape and width are unchanged and the mean (2.79) moves with it.

An open question

(slide 19–20)

This is deliberately left unresolved today.

Before moving on. How could you make each single reading more precise? Take a moment: what would you personally change?

Not answered today. We come back to it on 28 October, with the pendulum lab: physics has an answer you probably have not thought of.

Exam-style exercises

(slide 21–22)

Try them together. For a reason, be specific: “human error” on its own does not explain anything.

1. **Rolling ball** (figure 5). A student times a ball rolling down a ramp five times. Her results are 1.84 s, 1.79 s, 2.31 s, 1.82 s and 1.80 s.
 - (a) Identify the anomalous result. [1]
 - (b) Calculate the mean time, ignoring the anomalous result. Give your answer to 3 significant figures. [2]
 - (c) Suggest one reason why her results are not all the same. [1]
2. **The balance that isn’t zero** (figure 6). With nothing on it, a digital balance reads 0.3 g. A student places a stone on the balance and it reads 25.6 g.
 - (a) State the name of this type of error. [1]
 - (b) Calculate the correct mass of the stone. [1]
 - (c) The student weighs the stone five times and takes the mean. Explain whether this removes the error. [1]

Solutions are at the end of the chapter.

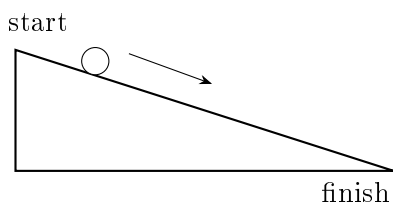


Figure 5: Exercise 1: a ball rolling down a ramp, timed from start to finish (schematic).

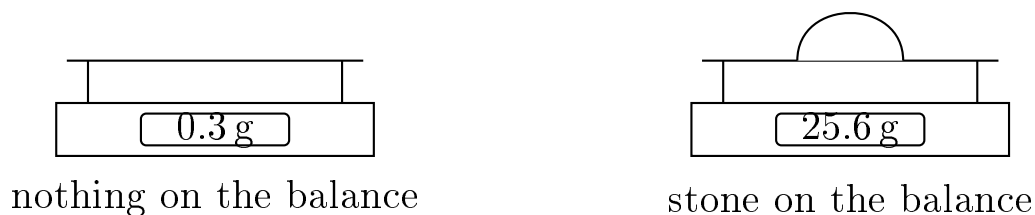


Figure 6: Exercise 2: the digital balance, empty and with the stone.

In summary

(slide 24)

- A single reading is limited by both the instrument’s sensitivity and human reaction time.
- Repeating a measurement and averaging reduces random error, not systematic error.
- An anomalous result needs a reason before it is excluded.
- Open question: how do we make a single measurement more precise? We come back to it on 28 October.

Solutions of the exercises

(slide 21–22)

1. (a) The anomalous result is 2.31 s. (b) Without the anomaly there are four readings: $1.84 + 1.79 + 1.82 + 1.80 = 7.25$ s. Mean = $7.25 \div 4 = 1.8125$ s, that is 1.81 s to 3 significant figures. (c) Her reaction time varies, at the start and/or at the stop; or the ball is released slightly differently each time.
2. (a) Zero error, which is a systematic error. (b) $25.6 - 0.3 = 25.3$ g. (c) No: every reading is 0.3 g too high, so the same shift remains in the mean.